

FRACTIONS

Short Revision Notes | Apply Level (Use Knowledge in New Situations)

1. Applying Fraction Operations

Operation	Rule	Example
Addition (same denominator)	Add numerators, keep denominator	$\frac{2}{7} + \frac{3}{7} = \frac{5}{7}$
Addition (different denominators)	Find LCM, convert, then add	$\frac{1}{2} + \frac{1}{3} = \frac{3}{6} + \frac{2}{6} = \frac{5}{6}$
Subtraction (same denominator)	Subtract numerators, keep denominator	$\frac{5}{6} - \frac{2}{3} = \frac{5}{6} - \frac{4}{6} = \frac{1}{6}$
Subtraction (different denominators)	Find LCM, convert, then subtract	$\frac{3}{4} - \frac{2}{5} = \frac{15}{20} - \frac{8}{20} = \frac{7}{20}$
Multiplication	Multiply numerators, multiply denominators	$\frac{2}{3} \times \frac{3}{4} = \frac{6}{12} = \frac{1}{2}$
Division	Keep, Change, Flip (multiply by reciprocal)	$\frac{2}{3} \div \frac{3}{4} = \frac{2}{3} \times \frac{4}{3} = \frac{8}{9}$

2. Applying "OF" Problems

Key Rule

- Replace "of" with "x" (multiplication)

Problem	Calculation	Answer
$\frac{1}{2}$ of 64 rupees	$\frac{1}{2} \times 64$	32 rupees
$\frac{2}{3}$ of 400 g	$\frac{2}{3} \times 400$	266 $\frac{2}{3}$ g
$\frac{1}{3}$ of 6 hectares	$\frac{1}{3} \times 6$	2 hectares
$\frac{1}{8}$ of 1 km	$\frac{1}{8} \times 1000$ m	125 m
$\frac{3}{4}$ of 500 m	$\frac{3}{4} \times 500$	375 m
$\frac{2}{5}$ of 1000 kg	$\frac{2}{5} \times 1000$	400 kg

Method:

Steps

- Replace "of" with "x"
- Multiply as fractions
- Simplify if needed

3. Applying Mixed Number Operations

Key Step

- Always convert mixed numbers to improper fractions FIRST!

Mixed Number Addition

Problem	Convert	Calculate	Answer
$1\frac{1}{2} + 2\frac{1}{4}$	$\frac{3}{2} + \frac{9}{4}$	$\frac{6}{4} + \frac{9}{4} = \frac{15}{4}$	$3\frac{3}{4}$
$2\frac{1}{3} + 1\frac{1}{2}$	$\frac{7}{3} + \frac{3}{2}$	$\frac{14}{6} + \frac{9}{6} = \frac{23}{6}$	$3\frac{5}{6}$
$3\frac{2}{5} + 1\frac{1}{3}$	$\frac{17}{5} + \frac{4}{3}$	$\frac{51}{15} + \frac{20}{15} = \frac{71}{15}$	$4\frac{11}{15}$

Mixed Number Subtraction

Problem	Convert	Calculate	Answer
$3\frac{5}{6} - 1\frac{2}{3}$	$\frac{23}{6} - \frac{5}{3}$	$\frac{23}{6} - \frac{10}{6} = \frac{13}{6}$	$2\frac{1}{6}$
$1\frac{1}{2} - \frac{2}{3}$	$\frac{3}{2} - \frac{2}{3}$	$\frac{9}{6} - \frac{4}{6} = \frac{5}{6}$	$\frac{5}{6}$
$4\frac{3}{4} - 2\frac{1}{5}$	$\frac{19}{4} - \frac{11}{5}$	$\frac{95}{20} - \frac{44}{20} = \frac{51}{20}$	$2\frac{11}{20}$

Mixed Number Multiplication

Problem	Convert	Calculate	Answer
$1\frac{1}{2} \times 2\frac{1}{4}$	$\frac{3}{2} \times \frac{9}{4}$	$\frac{27}{8}$	$3\frac{3}{8}$
$2\frac{1}{3} \times 1\frac{1}{2}$	$\frac{7}{3} \times \frac{3}{2}$	$\frac{21}{6} = \frac{7}{2}$	$3\frac{1}{2}$
$3\frac{2}{5} \times 1\frac{3}{4}$	$\frac{17}{5} \times \frac{7}{4}$	$\frac{119}{20}$	$5\frac{19}{20}$

Mixed Number Division

Problem	Convert	Keep, Change, Flip	Answer
$1\frac{1}{2} \div 2\frac{1}{4}$	$\frac{3}{2} \div \frac{9}{4}$	$\frac{3}{2} \times \frac{4}{9} = \frac{12}{18}$	$\frac{2}{3}$
$2\frac{1}{3} \div 1\frac{1}{2}$	$\frac{7}{3} \div \frac{3}{2}$	$\frac{7}{3} \times \frac{2}{3} = \frac{14}{9}$	$1\frac{5}{9}$
$3\frac{3}{4} \div 1\frac{1}{4}$	$\frac{15}{4} \div \frac{5}{4}$	$\frac{15}{4} \times \frac{4}{5} = \frac{60}{20}$	3

4. Applying BODMAS with Fractions

BODMAS Order

● Brackets -> Of -> Division -> Multiplication -> Addition -> Subtraction

Expression	BODMAS Application	Answer
$1/2 + 1/3 \times 1/4$	$1/3 \times 1/4 = 1/12$; $1/2 + 1/12 = 6/12 + 1/12 = 7/12$	$7/12$
$(1/2 + 1/3) \times 1/4$	$1/2 + 1/3 = 5/6$; $5/6 \times 1/4 = 5/24$	$5/24$
$2/3$ of $3/4 + 1/2$	$2/3 \times 3/4 = 1/2$; $1/2 + 1/2 = 1$	1
$1/2 + 2/3 \div 3/4$	$2/3 \div 3/4 = 2/3 \times 4/3 = 8/9$; $1/2 + 8/9 = 9/18 + 16/18 = 25/18$	$1 \frac{7}{18}$
$3/4 - 1/2$ of $1/3$	$1/2 \times 1/3 = 1/6$; $3/4 - 1/6 = 9/12 - 2/12 = 7/12$	$7/12$

More BODMAS Practice:

Expression	Step 1	Step 2	Step 3	Answer
$1/3$ of $(3/4 + 1/2)$	$3/4 + 1/2 = 5/4$	$1/3 \times 5/4$	-	$5/12$
$(2/3 \div 1/2) \times 1/4$	$2/3 \div 1/2 = 4/3$	$4/3 \times 1/4 = 4/12$	-	$1/3$
$1 \frac{1}{2} + 2 \frac{3}{4} \div 1/2$	$2 \frac{3}{4} \div 1/2 = 11/4 \times 2/1 = 11/2$	$3/2 + 11/2 = 14/2$	-	7
$(1 \frac{1}{2} + 2 \frac{3}{4}) \div 1/2$	$3/2 + 11/4 = 6/4 + 11/4 = 17/4$	$17/4 \div 1/2 = 17/4 \times 2/1 = 34/4$	-	$8 \frac{1}{2}$
$1/2 \times 3/4 \div 1/3$ of $1/4$	$1/3 \times 1/4 = 1/12$	$1/2 \times 3/4 = 3/8$	$3/8 \div 1/12 = 3/8 \times 12/1 = 36/8$	$4 \frac{1}{2}$

5. Applying Reciprocals

Key Rule

- To find a reciprocal, flip the numerator and denominator

Number	Reciprocal	Check (Product = 1)
$\frac{2}{3}$	$\frac{3}{2}$	$\frac{2}{3} \times \frac{3}{2} = 1$
$\frac{3}{4}$	$\frac{4}{3}$	$\frac{3}{4} \times \frac{4}{3} = 1$
$\frac{1}{5}$	5	$\frac{1}{5} \times 5 = 1$
3	$\frac{1}{3}$	$3 \times \frac{1}{3} = 1$
$2\frac{1}{2}$ ($\frac{5}{2}$)	$\frac{2}{5}$	$\frac{5}{2} \times \frac{2}{5} = 1$
$1\frac{3}{4}$ ($\frac{7}{4}$)	$\frac{4}{7}$	$\frac{7}{4} \times \frac{4}{7} = 1$
$2\frac{1}{3}$ ($\frac{7}{3}$)	$\frac{3}{7}$	$\frac{7}{3} \times \frac{3}{7} = 1$

Applying Reciprocals to Division:

Division Problem	Convert to Multiplication	Answer
$\frac{3}{4} \div \frac{5}{8}$	$\frac{3}{4} \times \frac{8}{5} = \frac{24}{20}$	$\frac{6}{5} = 1\frac{1}{5}$
$\frac{5}{6} \div \frac{2}{3}$	$\frac{5}{6} \times \frac{3}{2} = \frac{15}{12}$	$\frac{5}{4} = 1\frac{1}{4}$
$\frac{1}{3} \div \frac{1}{5}$	$\frac{1}{3} \times \frac{5}{1} = \frac{5}{3}$	$1\frac{2}{3}$
$8 \div \frac{4}{5}$	$8 \times \frac{5}{4} = \frac{40}{4}$	10

6. Applying Equivalent Fractions

Key Rule

- Multiply or divide both numerator and denominator by the SAME number

Finding Equivalent Fractions (Multiply)

Original	x2	x3	x4	x5
$\frac{1}{3}$	$\frac{2}{6}$	$\frac{3}{9}$	$\frac{4}{12}$	$\frac{5}{15}$
$\frac{3}{4}$	$\frac{6}{8}$	$\frac{9}{12}$	$\frac{12}{16}$	$\frac{15}{20}$
$\frac{2}{5}$	$\frac{4}{10}$	$\frac{6}{15}$	$\frac{8}{20}$	$\frac{10}{25}$

Simplifying Fractions (Divide)

Original	Divide by 2	Divide by 3	Divide by 4	Simplest Form
$\frac{4}{6}$	$\frac{2}{3}$	-	-	$\frac{2}{3}$
$\frac{6}{9}$	-	$\frac{2}{3}$	-	$\frac{2}{3}$
$\frac{8}{12}$	$\frac{4}{6}$	-	$\frac{2}{3}$	$\frac{2}{3}$
$\frac{12}{16}$	$\frac{6}{8}$	$\frac{4}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
$\frac{20}{25}$	-	-	$\frac{5}{5}$	$\frac{4}{5}$

7. Applying Fractions in Real-World Situations

Type 1: Finding a Fraction of an Amount

Problem	Method	Answer
Nimal's monthly income is Rs 40,000. He spends $\frac{1}{8}$ on travel. How much?	$\frac{1}{8} \times 40,000$	Rs 5,000
A person owns $\frac{3}{5}$ of a land. He gives $\frac{1}{3}$ to his daughter. What fraction does daughter get?	$\frac{3}{5} \times \frac{1}{3} = \frac{3}{15}$	$\frac{1}{5}$ of the whole land
A journey: walked $\frac{1}{8}$, train $\frac{2}{3}$, bus remaining. What fraction by bus?	$1 - (\frac{1}{8} + \frac{2}{3}) = 1 - (\frac{3}{24} + \frac{16}{24}) = 1 - \frac{19}{24}$	$\frac{5}{24}$

Type 2: Distribution Problems

Problem	Method	Answer
$\frac{1}{4}$ of income for food, $\frac{1}{2}$ for business. What fraction saved?	$1 - (\frac{1}{4} + \frac{1}{2}) = 1 - \frac{3}{4}$	$\frac{1}{4}$
Father gives $\frac{1}{2}$ land to son, $\frac{1}{3}$ to daughter. Son donates $\frac{1}{5}$ of his portion. What fraction of total land did son donate?	$\frac{1}{2} \times \frac{1}{5}$	$\frac{1}{10}$
Foundation builds on half the land it receives. What fraction of total land?	$\frac{1}{2} \times (\frac{1}{2} + \frac{1}{3}) \times \frac{1}{2}$	Calculate carefully

Type 3: Mixed Word Problems

Problem	Solution
A person puts aside $\frac{1}{4}$ of income for food and $\frac{1}{2}$ for business. What fraction does he save?	$1 - (\frac{1}{4} + \frac{1}{2}) = 1 - \frac{3}{4} = \frac{1}{4}$
Kumuduni walked $\frac{1}{8}$ of journey, travelled $\frac{2}{3}$ by train. What fraction by bus?	$1 - (\frac{1}{8} + \frac{2}{3}) = 1 - (\frac{3}{24} + \frac{16}{24}) = 1 - \frac{19}{24} = \frac{5}{24}$
Father gives $\frac{1}{2}$ land to son and $\frac{1}{3}$ to daughter. Son donates $\frac{1}{5}$ of his portion. What fraction did son donate?	$\frac{1}{2} \times \frac{1}{5} = \frac{1}{10}$

8. Applying "OF" to Different Units

Unit Type	Problem	Calculation	Answer
Money	$\frac{3}{4}$ of 64 rupees	$\frac{3}{4} \times 64 = 192/4$	48 rupees
Weight	$\frac{2}{5}$ of 400 g	$\frac{2}{5} \times 400 = 800/5$	160 g
Area	$\frac{1}{3}$ of 6 ha	$\frac{1}{3} \times 6 = 6/3$	2 ha
Length	$\frac{1}{8}$ of 1 km	$\frac{1}{8} \times 1000 = 1000/8$	125 m
Volume	$\frac{3}{5}$ of 50 L	$\frac{3}{5} \times 50 = 150/5$	30 L
Time	$\frac{3}{4}$ of 1 hour	$\frac{3}{4} \times 60 = 180/4$	45 minutes

9. Applying Fractions with Different Denominators

Problem	LCM	Convert	Answer
$\frac{2}{3} + \frac{1}{4}$	12	$\frac{8}{12} + \frac{3}{12}$	$\frac{11}{12}$
$\frac{3}{4} - \frac{1}{3}$	12	$\frac{9}{12} - \frac{4}{12}$	$\frac{5}{12}$
$\frac{5}{6} + \frac{5}{8}$	24	$\frac{20}{24} + \frac{15}{24}$	$\frac{35}{24} = 1 \frac{11}{24}$
$\frac{7}{8} - \frac{4}{5}$	40	$\frac{35}{40} - \frac{32}{40}$	$\frac{3}{40}$
$\frac{2}{3} + \frac{3}{4}$	12	$\frac{8}{12} + \frac{9}{12}$	$\frac{17}{12} = 1 \frac{5}{12}$
$\frac{4}{5} - \frac{1}{3}$	15	$\frac{12}{15} - \frac{5}{15}$	$\frac{7}{15}$

10. Applying BODMAS - Complex Problems

Expression	BODMAS Steps	Answer
$\frac{1}{4}$ of $(3\frac{1}{3} \div 2\frac{1}{6})$	$3\frac{1}{3} = \frac{10}{3}$, $2\frac{1}{6} = \frac{13}{6}$; $\frac{10}{3} \div \frac{13}{6} = \frac{10}{3} \times \frac{6}{13} = \frac{20}{13}$; $\frac{1}{4} \times \frac{20}{13} = \frac{20}{52} = \frac{5}{13}$	$\frac{5}{13}$
$(1\frac{2}{3} \times \frac{3}{5}) + (\frac{3}{4} + \frac{1}{2})$	$1\frac{2}{3} = \frac{5}{3}$; $\frac{5}{3} \times \frac{3}{5} = 1$; $\frac{3}{4} + \frac{1}{2} = \frac{5}{4}$; $1 + \frac{5}{4} = \frac{9}{4}$	$2\frac{1}{4}$
$2\frac{2}{3} \times (1\frac{1}{4} - \frac{1}{12}) \div 2\frac{1}{3}$	$1\frac{1}{4} - \frac{1}{12} = \frac{15}{12} - \frac{1}{12} = \frac{14}{12} = \frac{7}{6}$; $2\frac{2}{3} = \frac{8}{3}$; $\frac{8}{3} \times \frac{7}{6} = \frac{56}{18} = \frac{28}{9}$; $\frac{28}{9} \div \frac{7}{3} = \frac{28}{9} \times \frac{3}{7} = \frac{84}{63} = \frac{4}{3}$	$1\frac{1}{3}$
$\frac{5}{6} \div \frac{7}{18}$ of $\frac{2}{3} \times \frac{3}{4}$	$\frac{7}{18} \times \frac{2}{3} = \frac{14}{54} = \frac{7}{27}$; $\frac{5}{6} \div \frac{7}{27} = \frac{5}{6} \times \frac{27}{7} = \frac{135}{42} = \frac{45}{14} = 3\frac{3}{14}$; $\frac{45}{14} \times \frac{3}{4} = \frac{135}{56} = 2\frac{23}{56}$	$2\frac{23}{56}$

11. Applying Fractions to Word Problems - Step by Step

Step	Action	Example
1	Read problem carefully	A person owns $\frac{3}{5}$ of land, gives $\frac{1}{3}$ to daughter
2	Identify fractions and operations	$\frac{3}{5} \times \frac{1}{3}$
3	Solve	$\frac{3}{5} \times \frac{1}{3} = \frac{3}{15} = \frac{1}{5}$
4	Write answer with context	Daughter receives $\frac{1}{5}$ of the whole land

Word Problem Application Table

Type	Problem	Solution
Find a fraction of	What is $\frac{3}{4}$ of 64?	$\frac{3}{4} \times 64 = 48$
Find remaining fraction	Spent $\frac{1}{4}$ on food, $\frac{1}{2}$ on business. Saved?	$1 - (\frac{1}{4} + \frac{1}{2}) = 1 - \frac{3}{4} = \frac{1}{4}$
Find fraction of fraction	$\frac{2}{3}$ of $\frac{1}{5}$ of a land	$\frac{2}{3} \times \frac{1}{5} = \frac{2}{15}$
Find part of whole	$\frac{3}{5}$ of land, gives $\frac{1}{3}$	$\frac{3}{5} \times \frac{1}{3} = \frac{1}{5}$

12. Applying Complex Fraction Operations

Example 1: Adding and Subtracting Mixed Numbers

Problem	Convert	LCM	Answer
$1\frac{1}{2} + 2\frac{1}{4} - 1\frac{2}{3}$	$3/2 + 9/4 - 5/3$	$\text{LCM}(2,4,3) = 12$	$18/12 + 27/12 - 20/12 = 25/12 = 2\frac{1}{12}$

Example 2: Multiplication of Mixed Numbers

Problem	Convert	Multiply	Simplify	Answer
$3\frac{3}{10} \times 2\frac{1}{3} \times 4\frac{2}{7}$	$33/10 \times 7/3 \times 30/7$	$33 \times 7 \times 30 / 10 \times 3 \times 7 = 6930/210$	33	33

Example 3: Division with Mixed Numbers

Problem	Convert	KCF	Answer
$5\frac{1}{5} \div 6/7$	$26/5 \div 6/7$	$26/5 \times 7/6 = 182/30$	$91/15 = 6\frac{1}{15}$
$1\frac{1}{2} \div 2\frac{1}{4}$	$3/2 \div 9/4$	$3/2 \times 4/9 = 12/18$	$2/3$

13. Applying Reciprocals in Problems

Type	Problem	Solution
Find reciprocal	Reciprocal of $2\frac{3}{5}$	$2\frac{3}{5} = \frac{13}{5} \rightarrow \text{reciprocal} = \frac{5}{13}$
Use reciprocal	$6 \div \frac{3}{4}$	$6 \times \frac{4}{3} = 8$
Use reciprocal	$\frac{7}{9} \div \frac{2}{3}$	$\frac{7}{9} \times \frac{3}{2} = \frac{21}{18} = \frac{7}{6}$
Check reciprocal	Verify $\frac{2}{3}$ and $\frac{3}{2}$ are reciprocals	$\frac{2}{3} \times \frac{3}{2} = \frac{6}{6} = 1$ (correct)

14. Applying Cancellation Method (Cross-Cancelling)

Problem	Cancel	Multiply	Answer
$\frac{2}{5} \times \frac{10}{3}$	$10 \div 5 = 2$	$\frac{1}{5} \times \frac{10}{3} = \frac{1}{3} \times 2$	$\frac{2}{3}$
$\frac{3}{4} \times \frac{8}{9}$	3 with 9 ($\div 3$), 4 with 8 ($\div 4$)	$\frac{3}{4} \times \frac{8}{9} = \frac{1}{3} \times 2$	$\frac{2}{3}$
$\frac{5}{6} \times \frac{12}{25}$	5 with 25 ($\div 5$), 6 with 12 ($\div 6$)	$\frac{1}{5} \times \frac{2}{5} = \frac{2}{25}$	$\frac{2}{25}$
$\frac{7}{8} \times \frac{16}{21}$	7 with 21 ($\div 7$), 8 with 16 ($\div 8$)	$\frac{1}{3} \times \frac{2}{1} = \frac{2}{3}$	$\frac{2}{3}$

Cancellation Steps

- Look for common factors between numerator and denominator
- Cancel (divide) by those common factors
- Multiply remaining numbers
- Simplify if needed

15. Application Checklist

When adding/subtracting fractions:

- Find the LCM of denominators
- Convert each fraction to equivalent fraction
- Add/subtract numerators
- Simplify the answer

When multiplying fractions:

- Convert mixed numbers to improper fractions
- Cancel common factors (if possible)
- Multiply numerators, multiply denominators
- Convert back to mixed number if needed

When dividing fractions:

- Convert mixed numbers to improper fractions
- Keep the first fraction
- Change / to $\times$
- Flip the second fraction (reciprocal)
- Multiply and simplify

When using BODMAS:

- Brackets first
- "Of" means multiply
- Division and Multiplication (left to right)
- Addition and Subtraction (left to right)

When solving word problems:

- Read carefully
- Identify the fractions and operation
- Write the calculation
- Show your working
- Write the answer with units

16. Self-Test - Application Questions

No.	Question	Answer
1	Simplify: $\frac{2}{3} + \frac{1}{2}$	$\frac{7}{6} = 1 \frac{1}{6}$
2	Simplify: $\frac{3}{4} - \frac{1}{3}$	$\frac{5}{12}$
3	Simplify: $\frac{1}{2} \times \frac{2}{3}$	$\frac{1}{3}$
4	Simplify: $\frac{3}{4} \div \frac{3}{8}$	2
5	Find: $\frac{2}{3}$ of 120	80
6	Simplify: $1 \frac{1}{2} + 2 \frac{3}{4}$	$4 \frac{1}{4}$
7	Simplify: $1 \frac{1}{2} \div \frac{1}{3}$	$4 \frac{1}{2}$
8	Simplify: $(\frac{1}{2} + \frac{1}{3}) \times \frac{3}{4}$	$\frac{5}{8}$
9	Find reciprocal of $2 \frac{3}{4}$	$\frac{4}{11}$
10	Simplify: $\frac{2}{3}$ of $\frac{3}{4} \div \frac{1}{2}$	1

Revision Tip: Practice applying each operation regularly. Start with simple problems, then move to word problems. Always write your steps clearly — this helps you identify errors and shows your understanding!